

PROJECT REPORT ON USB PHYSICAL SECURITY

Submitted by

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COMPANY INTRODUCTION:

Supraja Technologies is a leading Knowledge and Technical Solutions Provider and pioneer leader in IT industry, is operating based out of Vijayawada.

R&D at Supraja

With a 24X7 work in Research & Development, experts at Supraja Technologies work under our :

- **Supraja Technologies Cyber Security Cell**

About Supraja Technologies:

Supraja Technologies (a unit of CHSMRLSS Technologies Pvt. Ltd.) with its foundation pillars as Innovation, Information and Intelligence is exploring indefinitely as a **Technology Service Provider (Corporate Consulting)** and as a **Training Organization (Ed-Tech)** as well.

You may visit us at :

www.suprajatechnologies.com

The multi domains of trainings which Supraja Technologies operate include the following :

- **Workshops & Hackathons**
 - Engineering Colleges
 - Corporate Companies (Startups & MNC's)
 - Government Organizations

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Branch Office : Prabhu Villas, Plot No. 124, 1st Floor, Church Road, Ayyappa Nagar, Vijayawada - 520007.

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- **Classroom Trainings Cum Certification Courses**

- Summer Training (30-45 Days)
- Winter Training (10 - 15 Days)
- Weekend Training (2 Days)
- 1 Month / 3 Months / 6 Months Courses

- **On-site Trainings**

- Value Added Courses / Two Credit Courses for Colleges
- Faculty Development Programs
- College Summer Training (15 Days, 30 Days, 45 Days & 60 Days)
- Govt Agencies, Police Academies, Corporates etc

- **Cloud Campus**

- (Distance Learning Program) *Coming Soon

- **Internships**

- Internship for Engineering Students (30 Days, 45 Days & 60 Days)
- Internship for Graduates (6 Months)

- **Lab Setup**

- Cyber Lab

- **CoE**

- Cyber Security Centre of Excellence

- **Supraja Technologies Security Assessment Product**

- (Our SaS Product is currently under development) *Coming Soon

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Why Supraja Technologies:

Be it Training or a workshop, the course content is always from R&D Cell of Supraja.

- A proven track record of delivering quality services.
- **68,500+** Students trained by our trainers till date.
- Training Partners of recognized institutions.
- Trainers with excellent research and teaching pedagogy illustrate their findings through corporate standard **practical demonstrations** during their sessions.
- Easy to learn and **hands-on sessions** are given, with additional benefits of Study Material, Tool kit and immediate query handling.
- Self-Prepared **Cyber Security Cell**.
- Supraja Technologies has the best, experienced and highly **skilled bunch of R&D Engineers, Security Analysts, Security Consultants & Trainers**.
- We provide training in Innovating and Trending Technologies to Govt. Officials, Corporate Houses and Colleges.

✓ Something we are proud of :

1. Supraja Technologies CEO Mr.Santosh Chaluvadi is an Alumni of Potti Sriramulu Chalavadi Mallikharjuna Rao College of Engineering and Technology, Vijayawada.

With our CEO this college conducted/organised a 50 hours Nonstop Marathon Training Workshop on Ethical Hacking & Cyber Security for which this respective college and our CEO both holds their name in **“LIMCA BOOK OF RECORDS 2017”**

2. We are very happy to inform you all that our company, Supraja Technologies has been shortlisted for **"Top 50 Tech Companies" award 2019**, conferred at InterCon - Dubai, UAE.

Supraja Technologies is one out of thousands companies that were initially screened by InterCon team of 45+ research analysts over a period of three months and the final shortlist includes 150+ firms and we are very proud to inform you all that our company Supraja Technologies also happens to be a part of the same.

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✓ Life changing solution/service :

After working on R&D for around 2 years, finally in the mid 2019 we have successfully developed a service/solution of various techniques and strategies for the Film Industry through which he can kill piracy of any film in online up to 35% right now. This betaservice is being appreciated & adopted by various Tollywood Film Industry Producers & Hero's to safeguard their film from piracy in online and to gain more profits.

By the end of 2033 our vision is to rollout a complete full packed service/solution where we can kill piracy entirely 100% everywhere in online for sure.

Appreciation :

Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri for providing our Anti-Piracy betaservice for his film VAJRA KAVACHADARA GOVINDA

✓ Achievements:

- On 17th August 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at Ramco Institute of Technology, Rajapalayam
- On 18th September 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at SRM University (Ramapuram Campus), Chennai
- On 20th November 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at St.Joseph's Institute of Technology, Chennai

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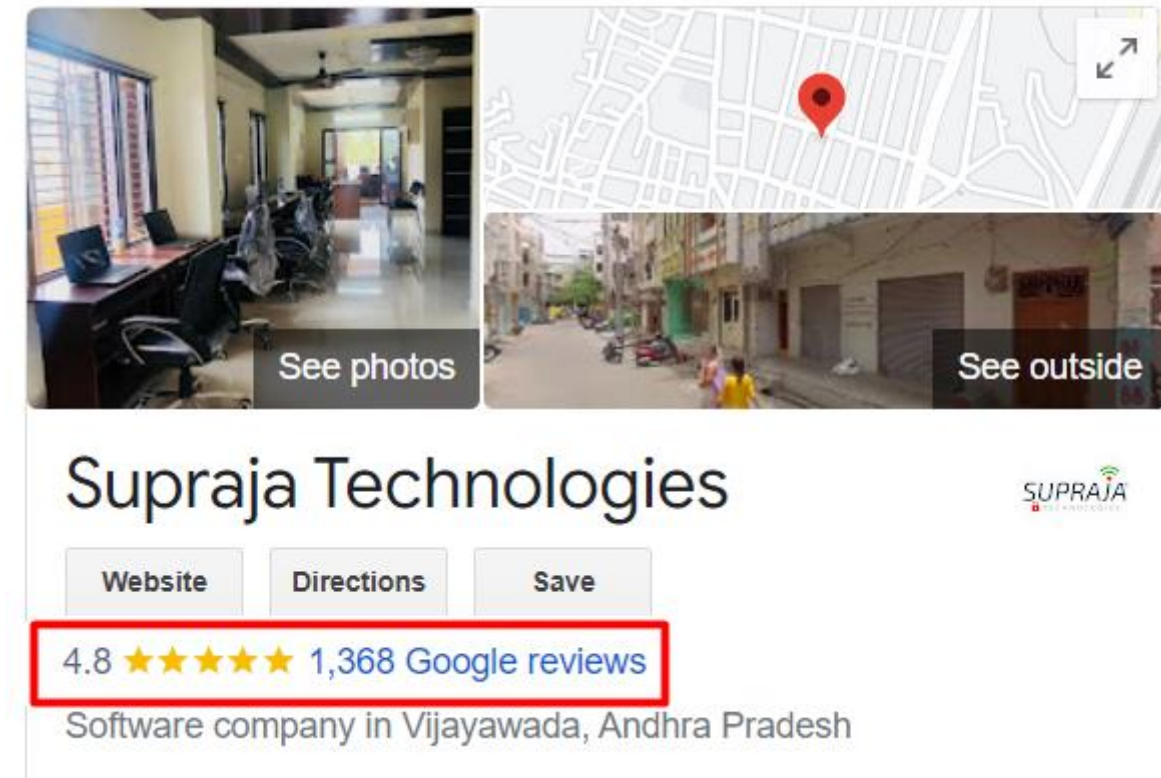
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**Our Company Supraja Technologies
is one of the emerging startup
in Andhra Pradesh
with **4.8** Google Ratings**



Link : <https://bit.ly/SuprajaGoogle>

AND

**also check our Company CEO Instagram Profile
for our recent more success stories:**

<https://www.instagram.com/chaluvadisantosh/>

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Santosh Chaluvadi

Founder & CEO
Supraja Technologies

He is a 28-year-old entrepreneur, one of the India's efficient Cyber Security Analyst and also he is an expert Digital Marketer as well. He is a digital marketer by profession and security enthusiast by passion. He primarily focuses on content building, testing and monetization of blogs. He has successfully developed many websites and done the security testing himself to ensure that the user's data is in safe hands and their privacy is protected. He is very active on social media and shares lot of tech stuff with his followers. The young student hacker has solved many issues with the vulnerabilities present in various websites and databases, given a solution in clearing the loopholes in order to protect the data to be leaked from the databases. Besides Ethical Hacking & Cyber Security, he also has a passion in Blogging & Digital Marketing.

While pursuing his engineering itself, he has trained many young generation people/students of more than 3500+ from various parts across Andhra Pradesh through his workshops, seminars, courses in Cyber Security and this makes him one of the youngest student trainers in India.

At the age of 20 he conducted his first workshop on Blogging & Ethical Hacking which was the beginning to his success in this field and right now he has handful of workshops to train students, government and corporate organizations as well in Andhra Pradesh & Telangana. He is the only student trainer who started conducting workshop for his peers, professors and for corporates.

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The year 2016 gave me the conviction and confidence to notch up whatever I was doing. I'd organized a 50-hour marathon event on Ethical Hacking and Cyber Security in PSCMR College, Vijayawada that went on to bag to achieve Limca Book of Records for non-stop longest duration workshop. The impact we were making was clearly visible by now. Diverse people from Jammu & Kashmir in the North to Kanyakumari in the South had attended the event. Some of the Government of India officials took notice of this record and invited our team for couple of conferences at New Delhi. The country's esteemed institutions were reaching out to me to conduct training, workshops, and events on a myriad of subjects related to online security. Multi-National Corporations (MNC's) had begun to consult me on the Cyber Security of their systems.

❖ Life changing solution/service :

After working on R&D for around 2 years, finally in the mid 2019 we have successfully developed a service/solution of various techniques and strategies for the Film Industry through which he can kill piracy of any film in online up to 35% right now. This betaservice is being appreciated & adopted by various Tollywood Film Industry Producers & Hero's to safeguard their film from piracy in online and to gain more profits.

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❖ Records, Appreciations, Awards & Recognitions etc at a glance :

1. Holds a National Record in “Limca Book of Records – 2017”
2. Ex-Associate Member for National Cyber Safety and Security Standards (NCSSS)
3. Steering Committee Member for United Conference on Cyber Space (UNITEDCON 2020)
4. Judge for the Grand Finale of SIH (Smart India Hackathon 2024) Software Edition for the Nodal Centre at Sri Sai Ram Engineering College, Chennai which is an initiative by Ministry of Education (Govt. of India) and AICTE
5. Received Appreciation from Mr.Amit Narayan, Executive Director at Rajahmundry Asset of Oil and Natural Gas Corporation Limited (ONGC) on 16th December, 2022 for training their employees on Cyber Security
6. Awarded as a "Karmaveer Chakra - 2019", on 12th October 2019 at IIT Delhi, which was instituted by iCONGO in partnership with the United Nations
7. Received Appreciation from Mr.Sandeep Rathore, Commissioner of Police, Chennai on 2nd March 2024 for the exceptional commitment and invaluable contribution as a Member of JURY at Greater Chennai Police Cyber Hackathon 3.0
8. Evaluator for the “Innovative Bharat” which is organized by AICTE and Ministry of Education held on 6th January 2024
9. Judge for the Grand Finale of SIH (Smart India Hackathon 2023) Senior Software Edition for the Nodal Centre at PVPSIT, Vijayawada which is an initiative by Ministry of Education (Govt. of India) and AICTE
10. Appointed as a Member for Board of Studies on 19th April 2025 for the Departments of Cyber Security, Data Science and AIML at Bapatla Engineering College, Bapatla
11. Appointed as a Member for Board of Studies on 12th March 2025 for the Department of Cyber Security at St. Joseph’s Institute of Technology, Chennai
12. Appointed as one of the Industry Academia Advisory Council Member on 30th September 2023 for the Department of Cyber Security at VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad
13. Appointed as a Member for Board of Studies on 4th May 2023 for the Department of Cyber Security at Madanapalle Institute of Technology & Sciences, Madanapalle
14. Appointed as a Member for Board of Studies for the Department of MCA at NRI Institute of Technology, Perecherla (Guntur)
15. Awarded as a “Social Media Influencer - 2019”, on 30th June 2019 by Jignasa in association with Government of Andhra Pradesh
16. Nominated for “INDIA 500 CEO AWARD 2019”
17. Invited & Interviewed by ETV Andhra Pradesh news channel on 27th July, 2019 for a Special Story Interview on “Spy Apps”
18. Appreciated by Mr.Sridhar, Sub-Inspector of Police at Central Crime Branch, Vijayawada on 23rd October, 2018 for exclusively training him on Special Investigation Course, which will help him to solve the cyber crime cases easily
19. Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri, for providing Anti-Piracy betaservice for his movie VAJRA KAVACHADARA GOVINDA

❖ Something we are proud of :

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Supraja Technologies is one out of thousands of startup companies that were initially screened by InterCon team of 45+ research analysts over a period of three months and the final shortlist includes 150+ firms and we are very proud to inform you all that our company Supraja Technologies also happens to be a part of the same.

Some Glimpses of our Journey



Mr.Santosh Chaluvadi – CEO, Supraja Technologies
Giving hands-on Cyber Security training workshop to the CSE students
at IIT Kharagpur

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Mr.Santosh Chaluvadi – CEO, Supraja Technologies was
Invited & Interviewed by ETV Andhra Pradesh news channel on 27th July, 2019
for a Special Story Interview on “Spy Apps”



Mr.Santosh Chaluvadi – CEO, Supraja Technologies was awarded as a
"Social Media Influencer 2019" in recognition of his remarkable achievements in the social
media as a part of First International Social Media Festival on 30th June 2019 by Jignasa in
association with Government of Andhra Pradesh

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On 23rd October 2018 Mr.Santosh Chaluvadi, CEO - Supraja Technologies and Mr.Krishna Chaitanya, CTO - Supraja Technologies had successfully completed delivering Special Investigation Course training in Cyber Security to Mr.Sridhar Garu, Sub-Inspector of Police at Central Crime Branch, Vijayawada which will help him to solve the cyber crime cases easily



ETV Andhra Pradesh News Channel interviewed Mr.Santosh Chaluvadi, CEO - Supraja Technologies for his achievements in the domain of Cyber Security & Digital Marketing

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Supraja Technologies was invited by Indian Air Force (Air Wing NCC) to deliver a session on Latest Cyber Crimes & Awareness for the NCC cadets, staff and officers on 4th July, 2019



Supraja Technologies – CEO, CTO & CMO with Indian Air Force (Air Wing NCC) Group Captain Sandeep Gupta. We thank Mr.Sandeep Gupta for inviting us on 4th July, 2019 to deliver a session on Latest Cyber Crimes & Awareness for the Indian Air Force (Air Wing NCC) cadets, staff & officers

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1. Introduction

1.1 Overall Description

Universal Serial Bus (USB) devices such as pen drives, external hard disks, and portable media have become an integral part of data storage and transfer in almost every organization and personal computer. While USB devices offer convenience and portability, they also introduce significant security risks. Unauthorized usage of USB ports can lead to data theft, malware injection, and leakage of sensitive information. In many enterprises, insider threats and external attacks often exploit unsecured USB ports to bypass traditional network-based security mechanisms. Therefore, it becomes essential to implement an additional layer of physical and logical security to control access to USB ports.

The project “USB Physical Security” aims to address this challenge by developing a system that enforces OTP-based access control for USB usage. The system ensures that before enabling or disabling USB ports, the user must authenticate themselves using a six-digit One-Time Password (OTP) sent to their registered email address. This mechanism guarantees that only authorized users can manage USB access. In addition, the project integrates intrusion detection measures — if an incorrect OTP is entered, the system automatically triggers the computer’s webcam to record a short video of the user attempting the unauthorized action. This recorded evidence serves as a strong deterrent against misuse and provides a reliable audit trail for administrators.

The system also incorporates logging and reporting functionality, where every activity such as OTP generation, USB enable/disable events, failed attempts, and intrusion recordings are stored in structured logs and reported to a backend server. These logs enhance accountability, allow centralized monitoring, and strengthen forensic investigation in case of security incidents.

Unlike existing systems that either permanently disable USB ports or rely on manual administrator intervention, this project provides a dynamic, user-friendly, and automated solution. The inclusion of a Graphical User Interface (GUI) using Python’s Tkinter library makes the application easy to operate, even for non-technical users. With a simple interface, users can request an OTP, allow USB access, or block USB ports with a single click, while the system handles all background security operations.

2. Existing System

In the current digital landscape, most computer systems and organizations rely on conventional methods to secure USB ports and external storage devices. The **existing systems** for USB security typically fall into three categories: permanent disabling of ports, manual administrative policies, or use of generic endpoint protection tools. While these methods provide some level of security, they have significant limitations and drawbacks that fail to address modern threats effectively.

One of the most common approaches is the **permanent disabling of USB ports** using system registry modifications or Group Policy settings in Windows. Once disabled, these ports remain inaccessible to all users unless manually reconfigured by an administrator. Although this prevents unauthorized devices from connecting, it also restricts legitimate users from using USB devices for approved tasks such as data backup, software updates, or transferring essential files. This “all-or-nothing” strategy lacks flexibility, creates inconvenience for employees, and often leads to operational inefficiencies. Moreover, it does not provide any monitoring or logging features, which means administrators cannot detect or trace attempted unauthorized usage.

Another method in practice is **manual administrative intervention**, where system administrators enable or disable USB ports when required. In this setup, the responsibility of managing USB access lies solely with IT personnel. This creates several practical challenges. First, it requires continuous human involvement, which is time-consuming and not scalable in large organizations. Second, there is a risk of insider threats, as administrators with elevated privileges could misuse their access. Third, manual systems do not leave sufficient digital evidence or audit trails, making it difficult to investigate incidents or enforce accountability.

In addition, many organizations rely on **generic endpoint protection software** such as antivirus programs, firewalls, or data loss prevention (DLP) tools. While these tools can detect malware or suspicious data transfers once a USB device is connected, they generally operate at the application or network layer rather than the physical port layer. This means they cannot prevent the initial act of plugging in an unauthorized device. Furthermore, these solutions are often resource-intensive, require costly licensing, and may not be tailored to specifically control USB port usage.

The major **limitations of existing systems** can be summarized as follows:



1. **Lack of authentication mechanisms** – Once USB ports are enabled, any user with physical access to the system can connect devices without identity verification.
2. **No real-time monitoring or deterrence** – Unauthorized attempts are neither tracked nor deterred, making systems vulnerable to repeated attacks.
3. **Absence of evidence collection** – In the event of misuse, existing systems provide no proof of the perpetrator’s identity or actions.
4. **Operational inefficiency** – Legitimate users face difficulties in accessing USB ports when rigid disabling methods are applied.
5. **High dependency on IT staff or third-party tools** – Organizations either overburden administrators with manual work or rely on expensive endpoint security software.

Therefore, while existing systems attempt to mitigate USB-related risks, they do so in a **reactive and inflexible manner**. They fail to balance **security with usability**, lack **multi-factor authentication**, and do not integrate mechanisms for **intrusion detection or logging**. These limitations highlight the urgent need for an improved, automated, and user-friendly system such as the proposed **USB Physical Security project**, which addresses these gaps through OTP-based authentication, real-time monitoring, and evidence collection.

3. Proposed System

The **USB Physical Security** project introduces an innovative and robust solution to overcome the shortcomings of existing USB security methods. Instead of relying on rigid disabling policies or manual administrator intervention, the proposed system combines **multi-factor authentication, real-time monitoring, and automated evidence collection** to ensure both **security and usability**. The system is designed to prevent unauthorized USB access while providing a flexible mechanism for legitimate users to safely enable or disable USB ports when required.

The cornerstone of the proposed system is its **OTP-based authentication mechanism**. When a user wishes to enable or disable USB ports, they must first request a **six-digit One-Time Password (OTP)**. This OTP is automatically generated by the system and delivered to the user's registered email address through a secure email service. Only users who can retrieve and correctly enter the OTP are granted permission to manage USB access. This ensures that USB ports are never freely accessible, and every action is linked to a verifiable user identity. By introducing this additional authentication layer, the system effectively minimizes risks posed by unauthorized employees, external intruders, or malware attempting to exploit open USB ports.

To further strengthen security, the system incorporates a **real-time intrusion detection mechanism**. If an incorrect OTP is entered, the system immediately assumes that an unauthorized attempt is being made. In response, it activates the computer's **webcam to record a short video clip** of the user sitting in front of the machine. This recorded footage is securely stored and logged, serving as undeniable digital evidence of the unauthorized activity. This feature not only acts as a deterrent but also provides administrators with valuable information for forensic investigation and accountability.

Another important feature of the proposed system is its **logging and reporting capability**. Every significant event—whether it is OTP generation, email delivery, successful USB enabling, port disabling, failed OTP attempts, or recorded intrusions—is captured in a structured log file. These logs are then transmitted to a central backend server through an API for monitoring and auditing. This centralization allows administrators to maintain a complete history of USB-related activities across all connected systems, enabling quick detection of suspicious patterns and compliance with organizational security policies.

Unlike existing systems, the proposed project is also designed with **usability in mind**. The system features a **Graphical User Interface (GUI)** built with Python's Tkinter library, ensuring that non-technical users can interact with it effortlessly. The GUI provides intuitive buttons to request an OTP, enable USB ports, disable USB ports, or view the user's profile. The interface guides the user through each step, displays clear messages for successes and failures, and minimizes the possibility of errors during operation. This user-centric design ensures that security measures do not come at the cost of convenience or productivity.

From a technical standpoint, the system leverages existing tools and libraries such as **OpenCV** for webcam access, **smtplib** for secure email communication, **requests** for API-based logging, and **winreg** for modifying Windows registry values that control USB port functionality. By integrating these components seamlessly, the proposed solution achieves strong security without the need for costly third-party software or specialized hardware. It runs efficiently on Windows-based systems, making it suitable for both personal computers and enterprise networks.

The **advantages of the proposed system** can be summarized as follows:

1. **Enhanced Security** – OTP verification ensures that only authorized users can manage USB access.
2. **Intrusion Detection** – Wrong OTP attempts trigger webcam recording for evidence collection.
3. **Accountability & Logging** – All activities are logged and transmitted to a central monitoring system.
4. **User-Friendly Interface** – Simple GUI makes the system accessible even to non-technical users.
5. **Cost-Effective Solution** – Built with open-source libraries, requiring no additional hardware or expensive licenses.
6. **Dynamic and Flexible** – Unlike permanent disabling methods, USB access can be enabled or disabled as needed with proper authentication.

4. System Design

4.1 Feasibility Study

4.1.1 Economical Feasibility

Economic feasibility evaluates the cost-effectiveness of the proposed USB Physical Security system. Traditional USB control solutions often require expensive **endpoint security software**, licensing costs, or specialized hardware-based port blockers. These solutions are not only costly but also demand regular updates and renewals, making them less attractive for small and medium-sized organizations.

In contrast, the proposed system is built entirely using **open-source technologies** such as Python, Tkinter, OpenCV, and the Windows registry API. Since these tools are freely available, the development cost is negligible. The only additional requirement is a standard webcam, which is already present in most modern laptops and can be purchased at a low cost for desktops. Furthermore, email-based OTP delivery uses free services like Gmail SMTP, avoiding the need for a dedicated paid messaging system.

Operational costs are also minimal. Once deployed, the system requires little to no maintenance beyond occasional updates. Unlike commercial security suites that consume heavy system resources and demand yearly subscriptions, this solution runs lightweight and efficiently, consuming minimal system resources.

From an economic perspective, the project offers **maximum security at minimal cost**, making it highly suitable for individuals, small businesses, and even large enterprises seeking budget-friendly data protection. Thus, the project is economically feasible as it provides a **cost-effective, long-term solution** compared to existing market alternatives.

4.1.2 Technical Feasibility

Technical feasibility analyzes whether the system can be implemented using the available tools, hardware, and software. The USB Physical Security project is designed to be lightweight, flexible, and easily deployable on **Windows operating systems**. It primarily modifies Windows Registry keys to enable or disable USB ports, a well-documented and stable approach.



The project integrates multiple modules such as OTP generation, email delivery, webcam recording, and logging through APIs. Each module uses widely adopted and technically proven Python libraries. For example, **smtplib** ensures secure email communication, **OpenCV** handles real-time webcam video capture, **Tkinter** provides a stable GUI, and **requests** enables seamless interaction with backend servers. These technologies are mature, reliable, and have strong community support.

From a hardware perspective, the system requires only a basic computer with **2 GB RAM, 40 GB storage, and a webcam**. This ensures compatibility with most existing organizational infrastructure without requiring additional investment in specialized devices.

Another advantage is **cross-compatibility of modules**. The system can easily be extended to support multi-platform deployments in the future, including Linux and macOS, by replacing the Windows registry control with platform-specific equivalents.

Since the project uses simple algorithms for OTP generation and modular design, the risk of failure is very low. Therefore, the system is technically feasible, ensuring reliable performance, scalability, and integration with current organizational setups.

4.1.3 Social Feasibility

Social feasibility focuses on how well the system will be accepted by end-users and organizations. One of the major challenges in cybersecurity systems is **user resistance**. If a system is too complex or inconvenient, employees may attempt to bypass it, weakening its effectiveness.

The proposed USB Physical Security system addresses this by offering a **user-friendly interface** through Tkinter. All functions—requesting OTP, enabling USB, disabling USB, and viewing profiles—are accessible with simple button clicks. Error and success messages are clearly displayed, reducing confusion and increasing user confidence.

From an organizational perspective, this system balances **security and convenience**. Unlike existing rigid systems that permanently disable USB ports, this solution allows controlled and flexible access. Employees can still use USB devices for legitimate purposes, but only after passing OTP authentication. This reduces frustration and improves productivity while maintaining strong security.

The **intruder recording feature** also enhances social accountability. Employees are less likely to misuse USB ports knowing their actions could be recorded and reported. This creates a culture of responsibility and deterrence against malicious activities.

Overall, the system will be well accepted as it requires **no special training**, minimizes disruption to work processes, and simultaneously boosts confidence in organizational data protection. Thus, the project is socially feasible, ensuring smooth adoption and long-term acceptance by users.

4.2 Input and Output Design

4.2.1 Input Design

Input design is critical because it determines how users interact with the system. The primary input in the USB Physical Security system is the **OTP (One-Time Password)** entered by the user. The OTP is generated by the system and sent to the user's registered email, ensuring that only authorized individuals can proceed. This introduces a **multi-factor authentication approach**, where both system access and email verification are required.

The input is designed with **error handling** and **validation checks**. For instance, the system ensures that only six-digit numeric codes are accepted, preventing accidental errors or tampering. The input process is simple: a pop-up dialog requests the OTP, and users type it in a secure, masked field to protect privacy.

Additionally, user confirmation inputs are required in certain scenarios. For example, before disabling USB ports, the system prompts a **Yes/No confirmation**, ensuring that accidental inputs do not lead to unintended actions.

The GUI ensures that inputs are not overwhelming for the user. All necessary instructions are displayed in dialog boxes, reducing the possibility of confusion. By focusing on simplicity, accuracy, and validation, the input design makes the system both secure and user-friendly.

4.2.2 Objectives

The objectives of the input/output design in the proposed system are as follows:

1. **Error Prevention** – Ensure that invalid OTPs or wrong inputs are handled gracefully without crashing the system.

2. **Ease of Use** – Design simple and clear input prompts so that even non-technical users can operate the system.
3. **Security Enforcement** – Guarantee that only authenticated inputs (valid OTPs) can modify USB access.
4. **User Guidance** – Provide confirmation dialogs, warnings, and error messages to guide the user at each step.
5. **Data Accuracy** – Ensure logs accurately record each input event, preventing manipulation or loss of information.

By meeting these objectives, the system ensures that security mechanisms are enforced without compromising usability.

4.2.3 Output Design

Output design ensures that the results of system operations are presented in a clear and useful manner. The outputs in the USB Physical Security project include:

- **System Messages** – Notifications for successful OTP validation, USB port enabling/disabling, or error alerts for incorrect OTPs. These are displayed in user-friendly pop-ups.
- **Intruder Recording** – When incorrect inputs are provided, the output includes a **recorded webcam video** of the unauthorized user. This output is saved locally and logged for future review.
- **Activity Logs** – JSON-based logs record events such as OTP generation, USB changes, and intrusion attempts. These logs are transmitted to the backend server, providing centralized reporting.
- **GUI Feedback** – The interface dynamically updates with status messages, ensuring that the user is always aware of the outcome of their actions.

A good output should clearly convey information, confirm user actions, and ensure accountability. The proposed system achieves this through secure dialogs, error alerts, and recorded evidence for enhanced security and awareness.

5. Implementation

5.1 Module Description

5.1.1 Email OTP Verification

The **Email OTP Verification** module serves as the foundation of authentication in the USB Physical Security system. Its primary purpose is to ensure that only authorized users can enable or disable USB ports. When a user requests access, the system generates a random **six-digit One-Time Password (OTP)** using Python's random library. This OTP is unique, valid for a single session, and prevents replay attacks or unauthorized reuse.

The OTP is securely transmitted to the user's **registered email address** through the Simple Mail Transfer Protocol (SMTP). The system leverages Gmail's SMTP service, configured with application-specific credentials stored in an environment file (.env) for security. This ensures that sensitive details like email passwords are never hardcoded into the program.

Once the OTP is delivered, the user must input it into the system through a Tkinter dialog box. If the entered OTP matches the generated value, USB access is granted or restricted depending on the chosen operation. Incorrect OTP attempts are logged and treated as possible intrusions.

This module adds a **multi-factor authentication layer**, ensuring that possession of the physical system alone is not enough for access. A valid, verified email account is also required, which significantly enhances security. The OTP mechanism prevents unauthorized insiders, malware, or external users from bypassing the protection. Thus, the Email OTP Verification module is a critical defense mechanism that enforces strong access control and accountability in the system.

5.1.2 USB Port Enabling/Disabling

The **USB Port Enabling/Disabling** module provides the system's primary functionality: controlling access to physical USB storage devices. In Windows, USB port behavior is governed by specific registry keys, particularly the **USBSTOR service key**. By modifying the value of this key, the system can dynamically enable or disable USB ports.

When the user successfully authenticates using OTP, the system modifies the registry key to either allow or block access. A value of **3** enables USB ports, while a value of **4** disables them. This

registry manipulation is carried out using Python's **winreg** module, which provides programmatic access to the Windows registry.

The design ensures that unauthorized users cannot alter USB access since the script requires **administrator privileges**. If run without proper permissions, the system alerts the user and halts execution, thereby maintaining strict security enforcement.

Additionally, the module includes safety checks such as confirmation dialogs before disabling USB ports. This prevents accidental blocking, ensuring that users are fully aware of their actions. All changes made through this module are recorded in structured logs for future auditing.

This dynamic approach offers flexibility compared to existing systems where ports are either permanently disabled or left unprotected. Users can still utilize USB devices for legitimate tasks while unauthorized attempts are blocked. The USB Port Enabling/Disabling module, therefore, strikes a balance between **security and usability**, making the system practical for real-world deployment.

5.1.3 Intrusion Detection via Webcam

The **Intrusion Detection via Webcam** module enhances the system by introducing a deterrent and accountability mechanism. Its purpose is to identify and document unauthorized attempts to gain USB access. If a user enters an incorrect OTP, the system treats it as a possible security breach. In response, it activates the computer's **webcam** and records a short **five-second video** of the individual at the terminal.

This functionality is implemented using the **OpenCV library**, which provides robust real-time access to video streams. The module captures frames at a standard resolution and saves the recording in .avi format using the **XVID codec**. To avoid overwriting, it automatically assigns unique filenames to each recording.

The captured video serves two purposes: (1) as evidence of the unauthorized attempt, and (2) as a **psychological deterrent**. Users are less likely to attempt unauthorized access if they know their actions may be recorded.

The module includes error handling to account for cases where the webcam is unavailable or blocked. In such cases, an appropriate error message is logged. Additionally, recordings are



automatically linked to the corresponding OTP attempt in the logs, ensuring accountability and ease of investigation.

By integrating this module, the system goes beyond traditional USB blocking solutions. It not only prevents unauthorized access but also **actively monitors and documents intrusions**, providing a powerful layer of physical security and transparency.

5.1.4 Logging and Reporting

The **Logging and Reporting** module ensures that every action within the USB Physical Security system is **tracked, stored, and communicated** to administrators. Logging begins at the moment the application is launched and records events such as OTP generation, email delivery success or failure, USB enabling/disabling, incorrect OTP attempts, and video recording triggers.

Logs are maintained in a **JSON structure**, making them lightweight, human-readable, and easy to parse programmatically. For accountability, each log entry is timestamped and associated with the corresponding action. This ensures that administrators can reconstruct the sequence of events during an investigation.

Beyond local storage, logs are also transmitted to a **centralized backend server** using HTTP POST requests. This is achieved with Python's **requests library**, which securely sends structured data to a predefined API endpoint. Centralized reporting ensures that even if logs are deleted locally, administrators still maintain a complete, tamper-proof history of all USB-related activities.

This module enhances organizational transparency and compliance with security policies. It enables **real-time monitoring**, where suspicious activity such as repeated failed OTP attempts can trigger immediate investigation. Additionally, the logging system helps in forensic analysis by linking OTP attempts, USB access, and intrusion recordings together.

Overall, the Logging and Reporting module ensures that the system not only prevents unauthorized USB access but also provides the necessary **audit trail** to enforce accountability and organizational security policies.

5.1.5 GUI Controller

The **GUI Controller** module provides the user interface through which all other modules are accessed. It is developed using **Tkinter**, Python's standard library for graphical interfaces. The

design philosophy is simplicity and usability, ensuring that even non-technical users can operate the system without difficulty.

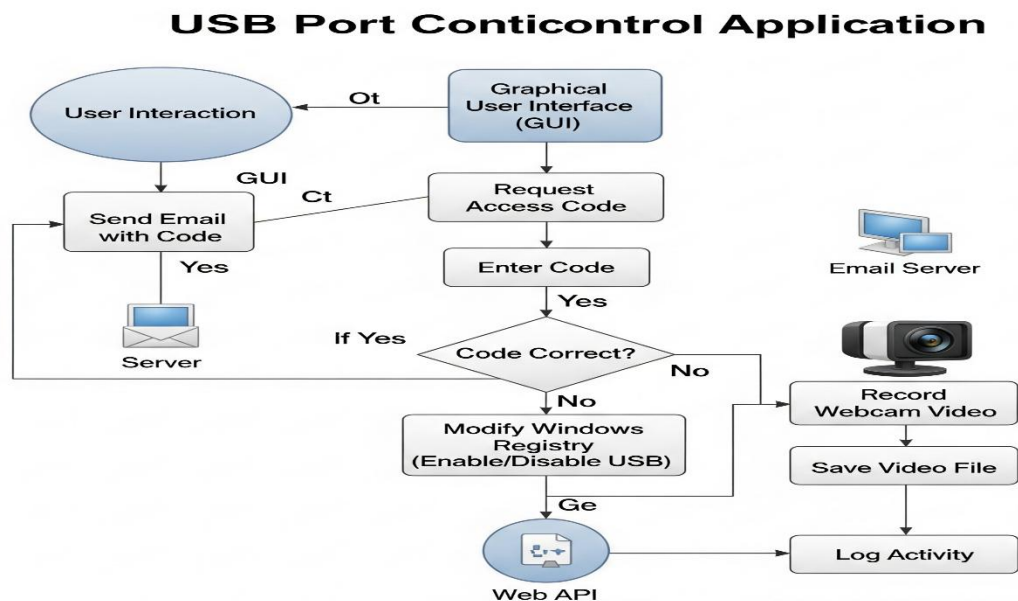
The main interface consists of buttons for critical actions: **Request Access Code**, **Allow USB Ports**, **Disable USB Ports**, and **View Profile**. Each button is linked to its respective backend function, enabling seamless interaction between the user and the underlying security modules. For example, clicking "Request Access Code" generates and emails an OTP, while "Allow USB Ports" verifies the OTP and modifies the registry accordingly.

The GUI also displays real-time feedback through **pop-up message boxes**. Success dialogs confirm completed actions, warning dialogs alert users to missing steps (such as forgetting to request an OTP), and error dialogs notify of incorrect OTP entries. This immediate feedback helps reduce confusion and errors during operation.

A key feature of the GUI Controller is that it enforces **workflow logic**. For example, users cannot enable or disable USB ports without first requesting and verifying an OTP. This ensures that security procedures are always followed in the correct sequence.

The GUI's intuitive design makes the system accessible while maintaining strict security. By abstracting the complexity of backend operations, the GUI Controller improves user experience and increases the likelihood of organizational adoption.

5.2 System Architecture



6. Algorithm Implementation

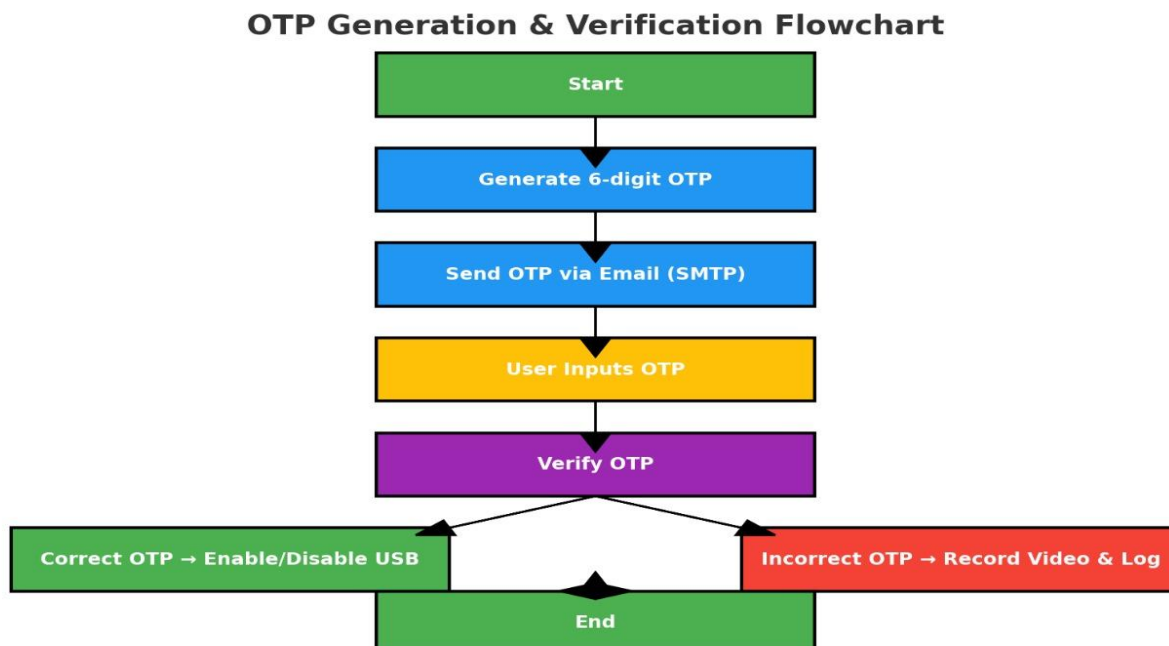
6.1 OTP Generation and Verification Logic

The core algorithm in the USB Physical Security system is the **One-Time Password (OTP) generation and verification logic**, which ensures that only authenticated users can enable or disable USB ports. It is designed to be lightweight, secure, and resistant to unauthorized reuse.

When a user requests access, the system generates a random **six-digit OTP** using Python's random library. The OTP is unique for that session and valid only for a limited duration. It is then sent securely to the user's registered email address via **SMTP**, ensuring that only individuals with access to the authorized email can retrieve the code.

Upon attempting to enable or disable USB ports, the system prompts the user to enter the received OTP. If the entered code matches the generated OTP, the request is validated, and the corresponding action is executed by modifying **Windows registry keys**. If the OTP does not match, the system assumes an unauthorized attempt and activates the **intrusion detection module**, which records a short webcam video and logs the incident.

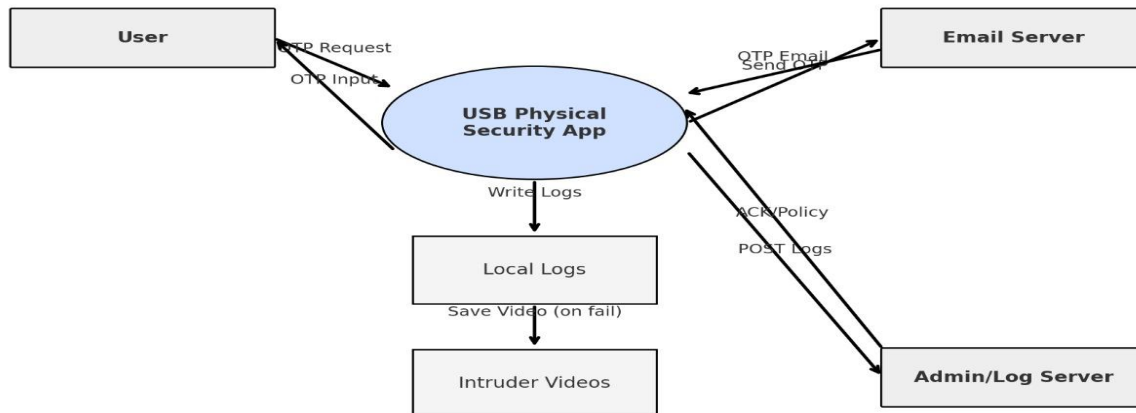
This OTP-based approach provides **multi-factor authentication**, enhances accountability, prevents misuse, and ensures that USB access remains **secure, monitored, and auditable**.



7. System Design

7.1 Data Flow Diagram (DFD)

7.1 Data Flow Diagram (Level-0) — USB Physical Security



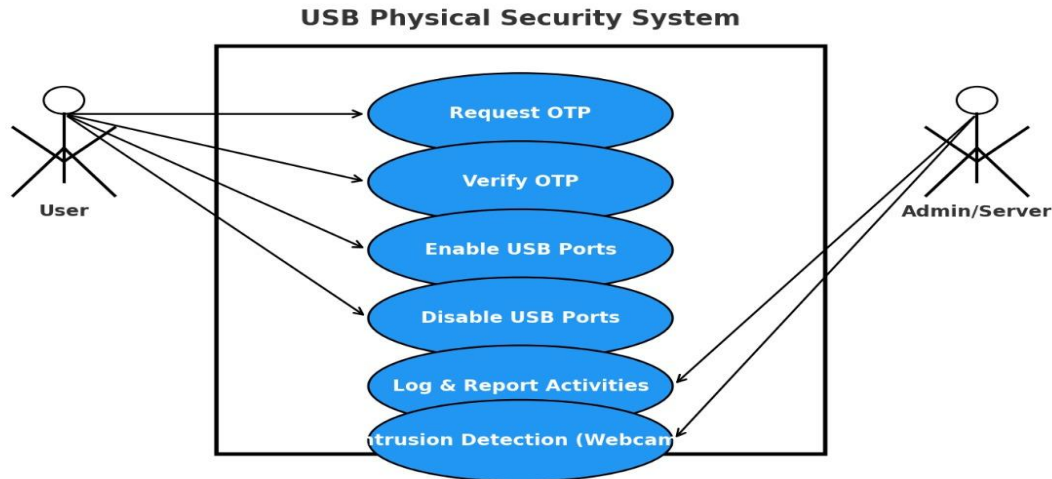
The Data Flow Diagram (DFD) of the USB Physical Security system illustrates the high-level flow of information between users, external entities, and system components. At the center of the diagram is the USB Physical Security Application, which manages OTP-based authentication, USB port control, intrusion detection, and logging.

The process begins with the User, who interacts with the system by requesting an OTP and entering it for verification. The application generates a unique OTP and forwards it to the Email Server, which delivers the OTP securely to the registered email account. Once the user enters the OTP, the application verifies it against the generated code.

If the OTP is valid, the system enforces USB access policies and records the event in Local Logs. If the OTP is invalid, the application activates the Intrusion Detector, saving a video of the unauthorized attempt under Intruder Videos. Both successful and failed attempts are written to Local Logs and reported to the Admin/Log Server for centralized monitoring and auditing.

This DFD highlights the secure interaction cycle, ensuring that every request is authenticated, every failure is monitored, and all events are logged, providing a robust layer of USB access security.

7.2 Use Case Diagram



The Use Case Diagram provides a high-level overview of how external actors (User and Admin/Server) interact with the USB Physical Security System. The User can perform activities such as requesting an OTP, verifying it, enabling USB ports, disabling USB ports, and viewing the results of these operations. Each of these actions ensures that only authorized users can manipulate USB access. Meanwhile, the Admin/Server has oversight responsibilities, such as monitoring logs, receiving reports, and verifying intrusion alerts.

This diagram is crucial because it defines the functional boundaries of the system. It makes it clear that the User is limited to authentication and USB access operations, while the Admin/Server has additional privileges related to system management and oversight. The separation of roles enhances security by ensuring accountability and restricting critical functions to authorized personnel.

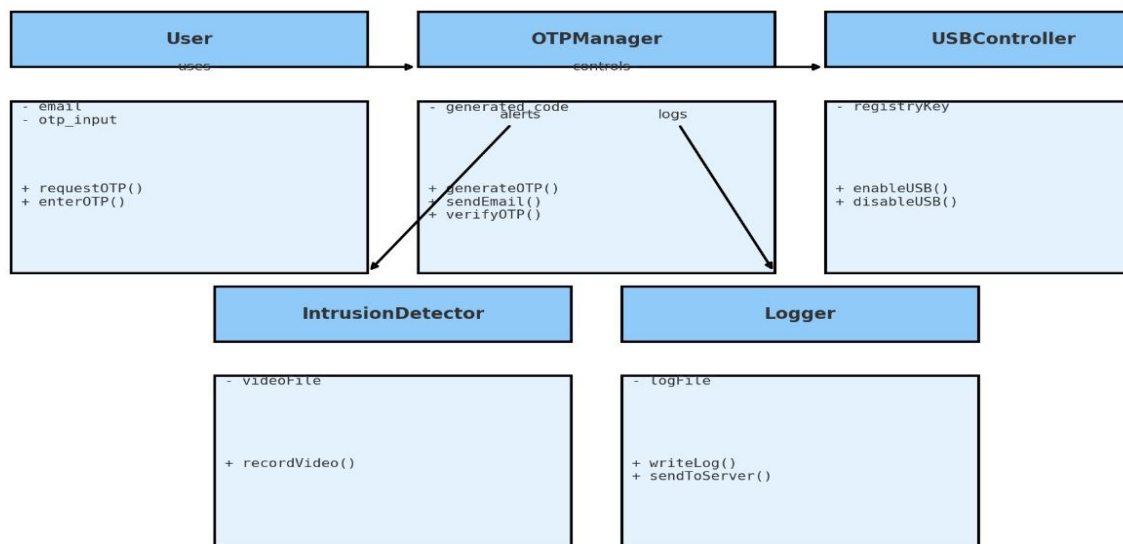
Furthermore, the use case diagram supports requirement traceability. Each use case corresponds to a functional requirement described in the specification. For instance, the “Verify OTP” use case directly aligns with the requirement that OTP authentication must be implemented for

enabling/disabling USB ports. The “Intrusion Detection” use case corresponds to the requirement for recording failed attempts.

Overall, the Use Case Diagram communicates the scope of the system to stakeholders in a simple, visual manner, ensuring clarity and alignment before moving into deeper system design.

7.3 Class Diagram

Class Diagram - USB Physical Security



The Class Diagram defines the structural blueprint of the USB Physical Security System by modeling its major classes, attributes, operations, and relationships. The diagram includes User, OTPManager, USBController, IntrusionDetector, and Logger as core classes.

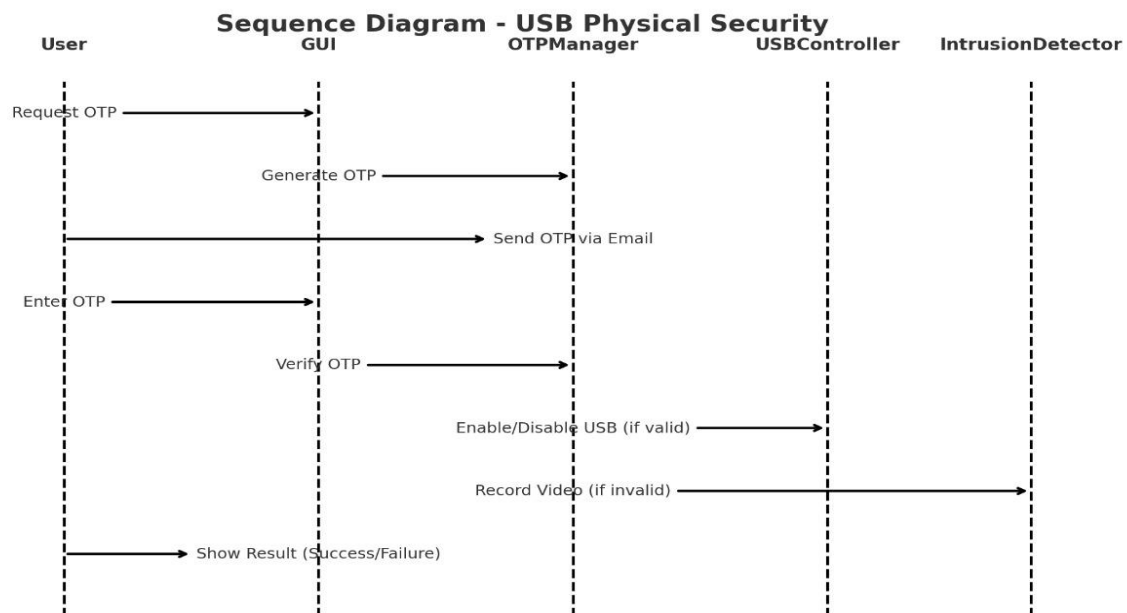
- The User class contains attributes such as email and OTP input. It defines operations like `requestOTP()` and `enterOTP()`, which represent the main interactions of the user with the system.
- The OTPManager class is responsible for OTP lifecycle management. It generates OTPs (`generateOTP()`), delivers them via email (`sendEmail()`), and verifies user inputs (`verifyOTP()`). Its relationship with User and Logger ensures secure communication and tracking.

- The USBController class contains attributes like registryKey and operations such as enableUSB() and disableUSB(). This directly manipulates Windows registry keys to enforce USB policies.
- The IntrusionDetector class is triggered during failed OTP verification attempts. It uses attributes like videoFile and operations such as recordVideo() to monitor unauthorized attempts.
- The Logger class maintains logs of all activities, including OTP events, USB access, and intrusion records. Its operations include writeLog() for local storage and sendToServer() for central reporting.

This class diagram promotes modularity and maintainability. Each class has a single responsibility, ensuring clean separation of concerns. It also illustrates dependencies—for example, OTPManager relies on Logger to record actions, and IntrusionDetector relies on OTPManager to determine failed attempts.

By defining system structure in this way, developers can implement and extend the system systematically. The diagram ensures that the software design follows object-oriented principles, making the solution scalable and easy to update.

7.4 Sequence Diagram



The Sequence Diagram models the dynamic interaction between components of the USB Physical Security System during a typical workflow. It visually demonstrates the order of messages exchanged among the User, GUI, OTPManager, USBController, and IntrusionDetector.

The sequence begins when the User requests an OTP via the GUI. This request is passed to the OTPManager, which generates a unique code using a random algorithm. The OTP is then sent to the registered email account. Once the user retrieves the OTP, they input it into the GUI. The GUI then forwards the input for verification.

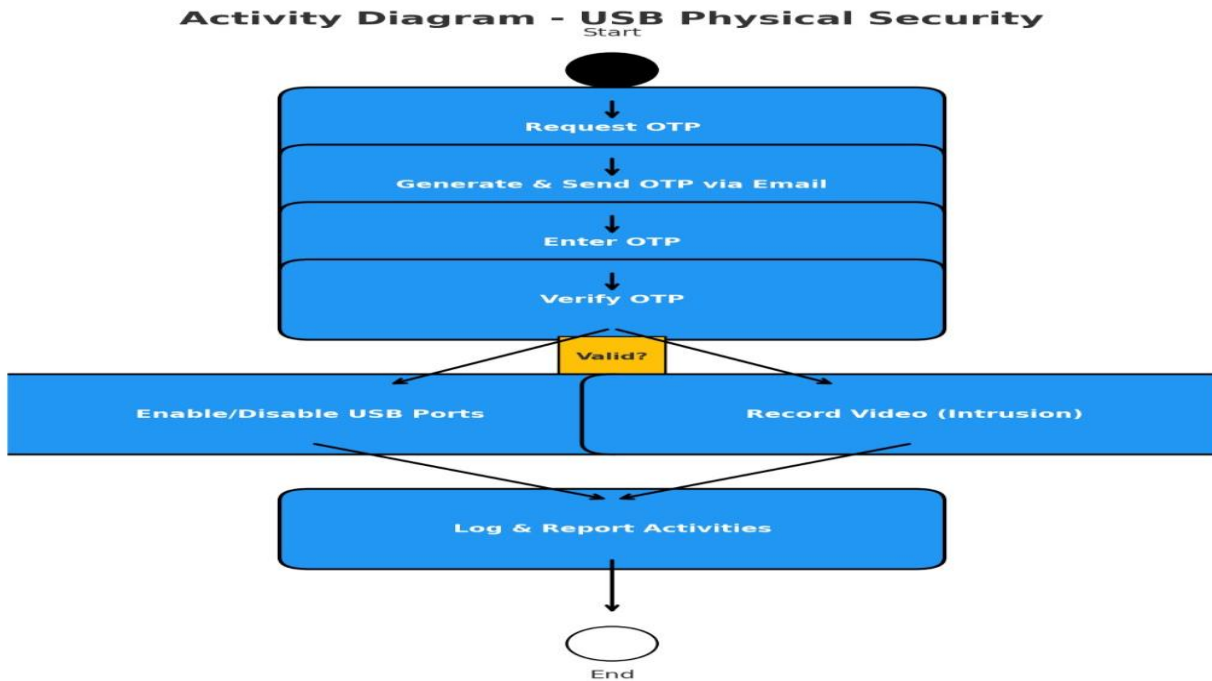
At this point, two scenarios can occur:

1. If the OTP is valid, the OTPManager confirms the match, and the USBController executes the requested action of enabling or disabling USB ports. The system then provides success feedback to the user.
2. If the OTP is invalid, the OTPManager signals failure, and the IntrusionDetector is triggered to record video evidence of the failed attempt. The system simultaneously logs the event for accountability and provides failure feedback to the user.

This sequence diagram is important because it captures real-time system behavior, showing how authentication, access control, and intrusion detection are tightly integrated. It ensures that stakeholders understand the flow of operations, while developers can use it as a reference to implement message-passing logic.

By modeling these interactions, the sequence diagram validates that the system enforces security rules consistently: only authenticated users gain access, while unauthorized attempts trigger monitoring and logging.

7.5 Activity Diagram



The **Activity Diagram** outlines the workflow of the USB Physical Security system from start to finish, focusing on the actions and decisions involved in granting or denying USB access. It begins with the user requesting an OTP. This request triggers the system to generate and send the OTP via email.

The user then enters the OTP, and the system proceeds to **verification**. At this stage, a decision node determines the validity of the OTP:

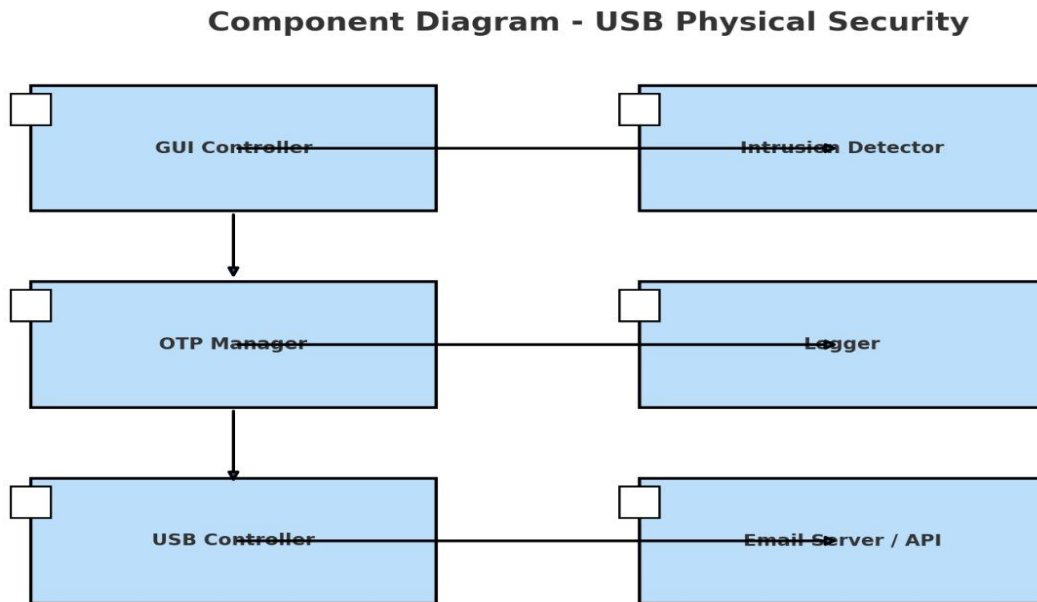
- **If valid**, the activity continues to enabling or disabling USB ports, depending on the user's request.
- **If invalid**, the system diverts to the intrusion detection path, where a webcam is activated to record video evidence of the unauthorized attempt.

Regardless of the branch taken, both outcomes converge at the final step: **log and report activities**. Logs are stored locally and forwarded to the server for auditing purposes.

This activity diagram is critical because it models **decision-making processes and parallel activities**. It ensures that the system consistently follows secure workflows, whether for successful

operations or failed attempts. Additionally, it highlights the **fail-safe design**—even failed actions lead to meaningful outcomes (intrusion recording and reporting).

7.6 Component Diagram



The Component Diagram presents the system at an architectural level, showing how major components interact to deliver USB security functionalities. Each component represents a modular unit with specific responsibilities.

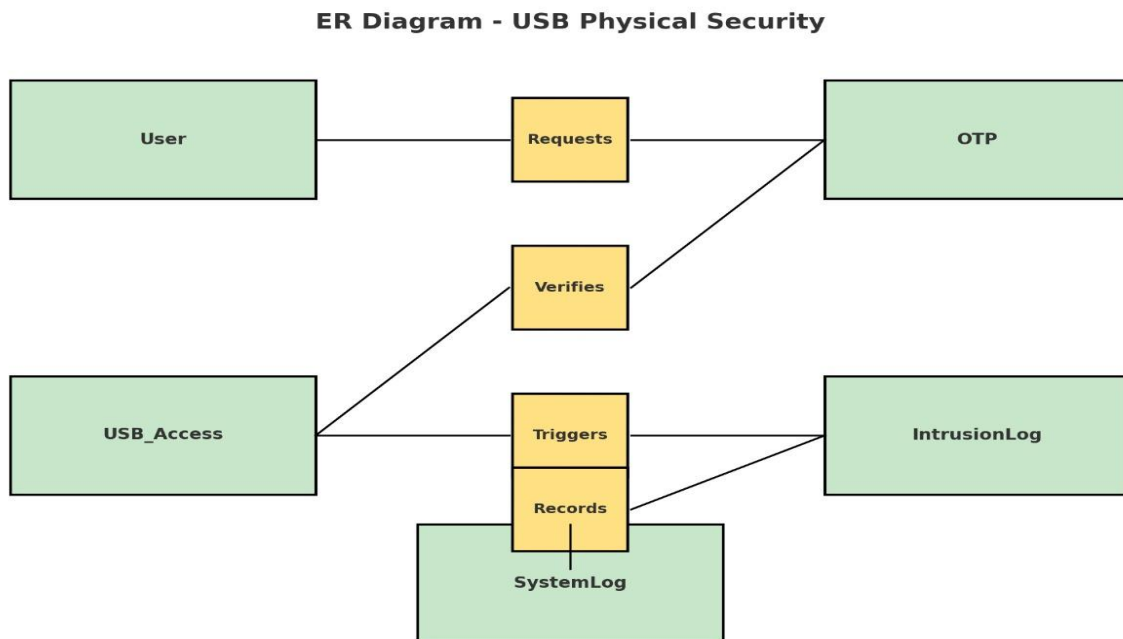
- The GUI Controller is the front-end interface where users interact with the system. It collects inputs such as OTP requests and verification codes.
- The OTP Manager handles the creation, delivery, and validation of OTPs. It communicates with the Email Server/API to send OTPs securely.
- The USB Controller enforces USB access policies by enabling or disabling ports through registry modifications.
- The Intrusion Detector is activated upon failed authentication attempts. It uses the system's webcam to record intrusions, creating video logs as evidence.

- The Logger maintains an auditable trail of all activities, including successful logins, failed attempts, and intrusion records. It can also forward data to the central log server for monitoring.

These components interact with minimal coupling, ensuring scalability and maintainability. For example, improvements in OTP generation logic won't affect the USB Controller, and intrusion detection enhancements won't disrupt OTP verification.

This diagram provides an architectural blueprint, showing that the system is built with modularity, security, and reliability in mind. It makes the design transparent for developers and easier to maintain for administrators.

7.7 ER-Diagram



The **Entity-Relationship Diagram** models the database design of the USB Physical Security system, highlighting entities, attributes, and relationships essential for secure operation.

Key entities include:

- **User:** Stores user details such as email and OTP input.
- **OTP:** Represents generated codes with attributes like code, timestamp, and status (valid/expired).



- **USB_Access:** Logs enabling or disabling events with details like time, user ID, and action performed.
- **IntrusionLog:** Stores data of unauthorized attempts, including video evidence.
- **SystemLog:** Consolidates all activities for reporting and auditing.

Relationships:

- A User can **request** multiple OTPs, and each OTP is linked to a single User.
- Each OTP is **verified** before granting access to USB_Access.
- USB_Access attempts can **trigger** IntrusionLog entries if invalid, linking access control with monitoring.
- Both USB_Access and IntrusionLog events are **recorded** in SystemLog for centralized auditing.

The ER diagram ensures that every authentication, authorization, and intrusion event is properly stored and traceable. It supports **data integrity, accountability, and auditability**. By clearly modeling relationships, it guarantees that administrators can monitor security events effectively.

8. Requirement Specification

8.1 Functional Requirements

The functional requirements of the USB Physical Security system define what the system must accomplish to provide strong protection against unauthorized USB access while ensuring usability for legitimate users. The primary functional requirements include:

1. OTP Generation and Delivery – The system must generate a unique six-digit OTP and deliver it securely to the registered email address of the user. This ensures only authenticated users can access the system.
2. OTP Verification – The system must validate the OTP entered by the user against the one generated. If valid, the system allows further actions; if invalid, intrusion detection is triggered.
3. USB Port Control – The system must enable or disable USB storage ports dynamically by modifying the Windows registry. This ensures flexibility without permanently locking or exposing ports.
4. Intrusion Detection – On failed OTP attempts, the system must capture a short video using the webcam for accountability.
5. Logging and Reporting – The system must record all activities such as OTP requests, USB access attempts, and intrusion alerts. Logs must be stored locally and transmitted to a central server for monitoring.
6. User Interface – The GUI must allow users to request OTP, enable/disable USB ports, and view status in an intuitive way.

These requirements ensure a multi-layered security mechanism with proper authentication, monitoring, and reporting capabilities.

8.2 Software Requirements

The system requires specific software components for development, execution, and integration with external services.

- Programming Language: Python (primary development language due to its simplicity, library support, and portability).
- Libraries and Frameworks:
 - tkinter – for GUI creation.
 - smtplib – for sending OTP via email.
 - random – for OTP generation.
 - winreg – for accessing and modifying Windows registry keys to control USB ports.
 - cv2 (OpenCV) – for webcam recording in intrusion detection.
 - requests – for API communication with central log server.
- Email Service: Gmail SMTP configured with an app password for secure OTP delivery.
- Database/Storage: JSON or flat file storage for local logs; centralized database for organizational deployment.
- IDE/Tools: Any Python IDE such as PyCharm, VS Code, or IDLE for development and debugging.
- Other Requirements: Administrator privileges on the system to modify USB registry keys.

These software requirements collectively ensure the smooth functioning of OTP-based verification, USB control, intrusion monitoring, and reporting.

8.3 Operating Systems Supported

The USB Physical Security system is designed primarily for Windows-based environments, as USB access control is implemented through modifications to the Windows Registry. Supported versions include:

- Windows 7 – Legacy systems where USB threats remain common.
- Windows 8 / 8.1 – Systems used in mid-generation environments.
- Windows 10 / 11 – Current mainstream operating systems in both enterprise and personal use.

Although the system is designed for Windows, its modular architecture makes it possible to extend to Linux-based operating systems in the future. For Linux, USB port access can be controlled by modifying system files (e.g., `/etc/fstab`) or through device manager policies. MacOS support is less common due to its closed architecture but may be achievable with custom extensions.

Thus, while the primary target OS is Windows, the project maintains flexibility to be adapted to other platforms. This focus ensures practicality since most enterprise and personal systems that face USB-related threats run on Windows.

8.4 Technologies and Languages Used to Develop

The project leverages widely used, reliable, and cost-effective technologies.

- Programming Language: Python (easy integration with system APIs, strong security libraries, and cross-platform support).
- GUI Development: Tkinter, chosen for its simplicity and native compatibility with Python.
- Email Communication: Gmail SMTP protocol with Python's `smtplib` to deliver OTPs securely.
- Intrusion Monitoring: OpenCV library for capturing real-time video evidence.
- System-Level Control: Winreg for registry modification to manage USB port access.
- Networking/Reporting: Requests library for communicating logs to a backend API server.

- Storage/Logging: JSON file format for local logs, with extension to central database servers.

By combining these technologies, the system ensures an integrated security solution. The reliance on open-source tools reduces costs and increases accessibility, making the project suitable for academic, organizational, and professional deployment.

8.5 Hardware Requirements

The hardware requirements of the USB Physical Security system are modest, ensuring the solution remains lightweight and deployable on most existing machines.

- Processor: Minimum Intel i3 or equivalent; recommended Intel i5/i7 for enterprise use.
- Memory (RAM): Minimum 2 GB; recommended 4 GB or higher for smooth functioning with video capture.
- Storage: At least 500 MB of free disk space for application files and logs.
- Webcam: Required for intrusion detection (can be inbuilt or external USB webcam).
- USB Ports: At least one USB 2.0/3.0 port to demonstrate enable/disable functionality.
- Internet Connection: Required for OTP email delivery and centralized log reporting.
- Permissions: The system must have administrator-level access on the host machine to allow registry modifications.

These requirements are intentionally kept minimal so that the system can run on both personal laptops and enterprise desktops without demanding additional infrastructure.

9. System Test

9.1 Types of Tests

9.1.1 Unit Testing

Unit testing was carried out on every independent module of the project. For example:

- **OTP Generation** was tested by running it 100 times to ensure uniqueness of codes and confirm that no OTPs repeated within a single session.
- **Email Delivery** was tested by simulating both valid and invalid SMTP credentials, ensuring that failure cases were properly logged.
- **USB Control Module** was tested by enabling/disabling ports multiple times to ensure system stability and rollback in case of failure.
- **Intrusion Detector** was tested by simulating failed logins and ensuring that video was captured for each attempt.

This level of testing ensured that **no hidden bugs** existed in isolated components.

9.1.2 Integration Testing

The focus of integration testing was on **data flow between modules**. For example:

- When OTP is generated, the Email Module must receive the OTP string correctly.
- When USB ports are disabled, the Logger must record the event in both local logs and server reports.
- If OTP verification fails, the Intrusion Detector must automatically start video recording.

Integration testing revealed issues like email latency and registry access delays, which were resolved by adding exception handling and retry logic.

9.1.3 Functional Testing

Functional testing was mapped directly to the **Requirement Specification (Section 8)**. Each requirement was turned into one or more test cases. Example test cases:

- **Requirement:** “System must disable USB ports after valid OTP input.”
 - *Test Case:* Enter valid OTP → Verify USB disabled.
- **Requirement:** “System must log unauthorized attempts.”
 - *Test Case:* Enter wrong OTP 3 times → Verify video recorded & log entry created.

This ensured **traceability between requirements and test cases**.

9.1.4 System Testing

System testing focused on **end-to-end workflows**. For example:

- Workflow 1: Request OTP → Receive Email → Enter OTP → Enable USB → Log Created.
- Workflow 2: Request OTP → Enter Wrong OTP → Video Captured → Unauthorized Log Sent.

Different system environments (Windows 7, 10, and 11) were used to confirm compatibility. Special emphasis was given to **security testing**, ensuring that OTPs could not be reused or bypassed.

9.1.5 White Box Testing

Developers performed **path coverage testing**, ensuring that every possible branch in OTP verification logic was executed:

- Valid OTP.
- Invalid OTP.
- Expired OTP.
- Multiple failed OTP attempts.

Exception handling was also tested, for example: “What if SMTP fails?” or “What if webcam not found?” This ensured internal code robustness.

9.1.6 Black Box Testing

Testers without access to the code tested the GUI by entering values into input fields and observing outputs. For example:

- Entering fewer than 6 digits in OTP field.
- Entering alphanumeric OTP instead of numeric.
- Requesting multiple OTPs simultaneously.

The results confirmed that the GUI properly displayed error messages and did not reveal internal system errors to the user (avoiding information leakage).

9.2 Test Strategy and Approach

The testing approach followed an **incremental model**:

1. **Unit Testing** of each module.
2. **Module Integration Testing** step by step.
3. **System Testing** on complete application.
4. **User Acceptance Testing** for deployment.

Both **manual and automated testing** were applied. Manual testing validated GUI interactions and system responses. Automated testing using Python scripts validated OTP uniqueness and registry changes.

Security was the **main focus of test strategy**. Negative testing (deliberately entering wrong data) was performed heavily to ensure that the system never compromised security.

9.3 Integration Testing

Key integrations tested:

- **GUI ↔ OTP Manager:** Verified that user requests trigger OTP generation and that entered OTP is correctly passed for validation.
- **OTP Manager ↔ Email Service:** Ensured successful OTP delivery and error logging in case of SMTP failure.



- **OTP Verification ↔ USB Controller:** Confirmed that only valid OTPs could change USB access.
- **OTP Verification ↔ Intrusion Detector:** Confirmed that failed attempts always triggered webcam recording.
- **Logger ↔ Central Server:** Ensured logs were correctly stored both locally and remotely.

This testing confirmed that **system modules work as a unified whole**.

9.4 Acceptance Testing

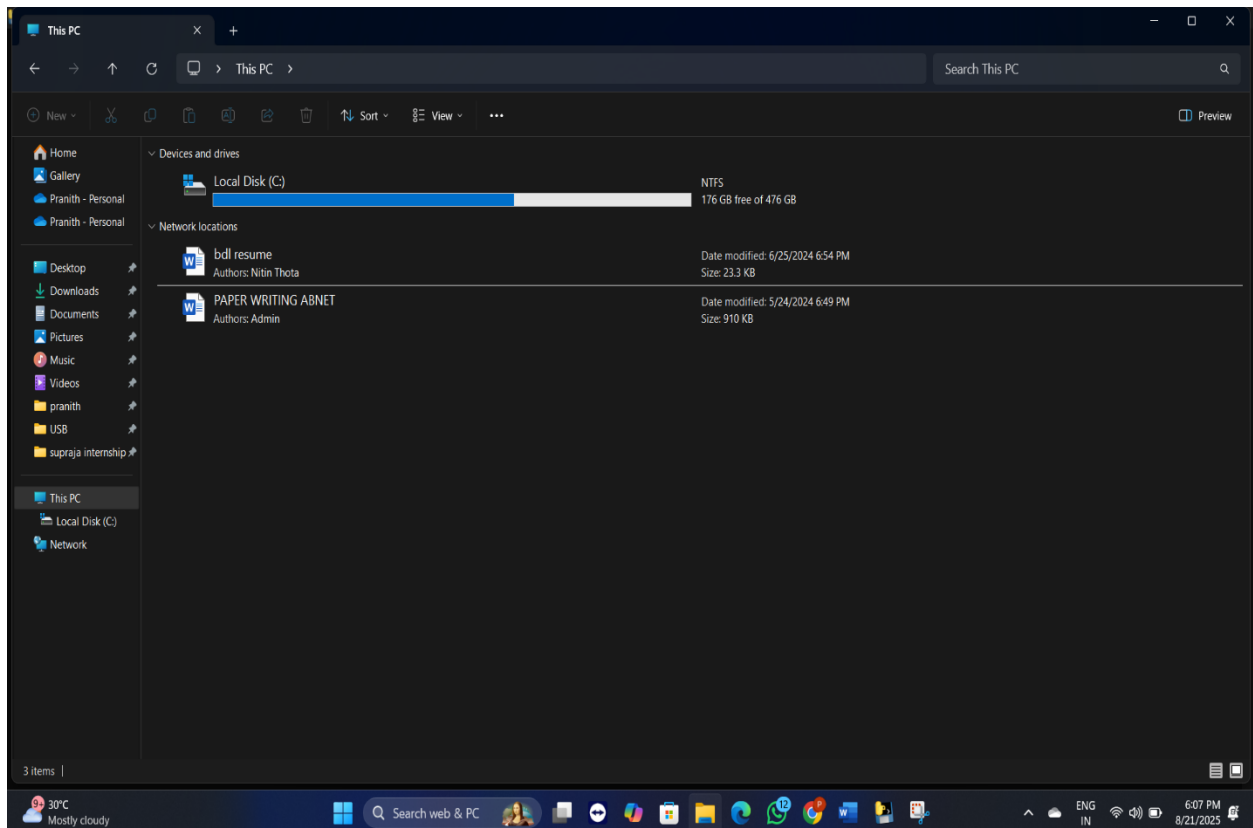
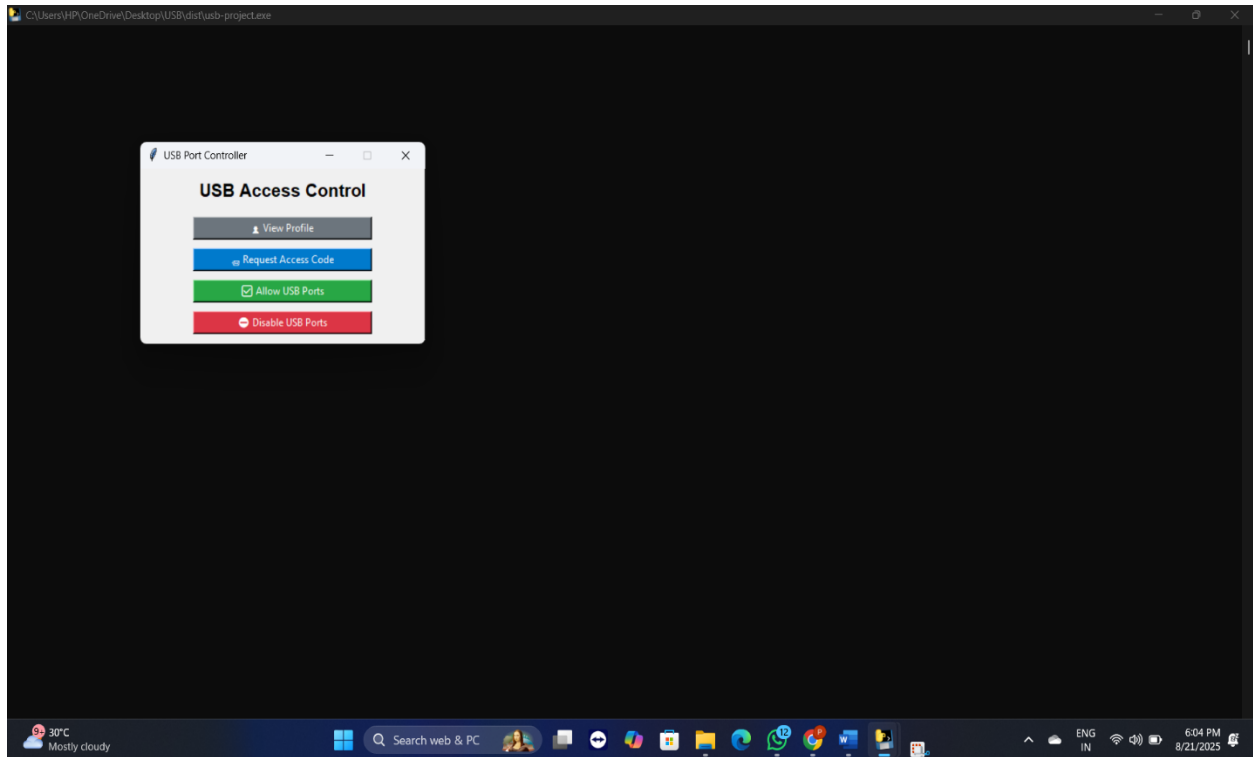
Acceptance testing was done in a **real-world setting**. A group of test users were asked to simulate both legitimate and malicious behaviors:

- **Scenario 1:** Legitimate employee enabling USB using valid OTP.
- **Scenario 2:** Unauthorized user attempting to guess OTP.
- **Scenario 3:** Multiple failed attempts to trigger intrusion detection.

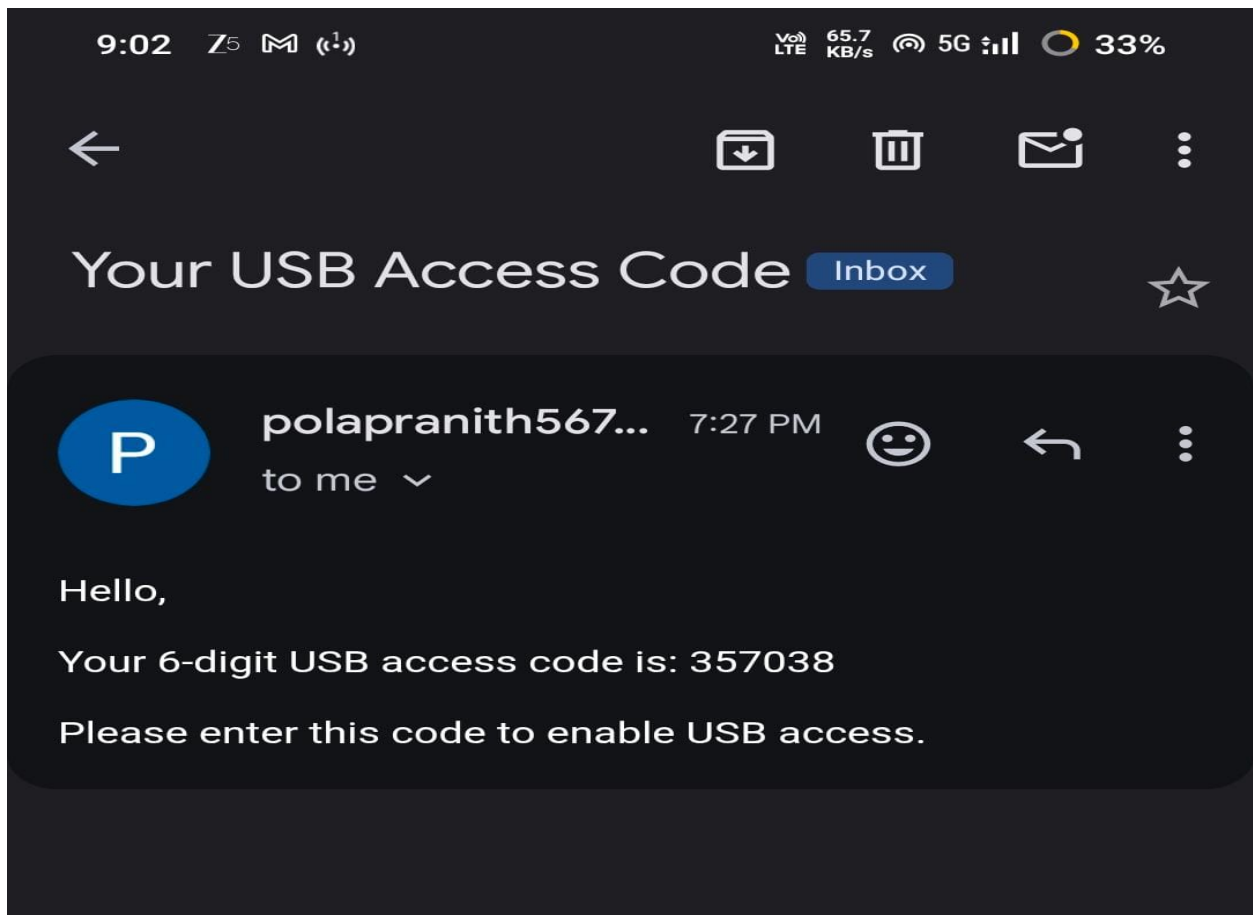
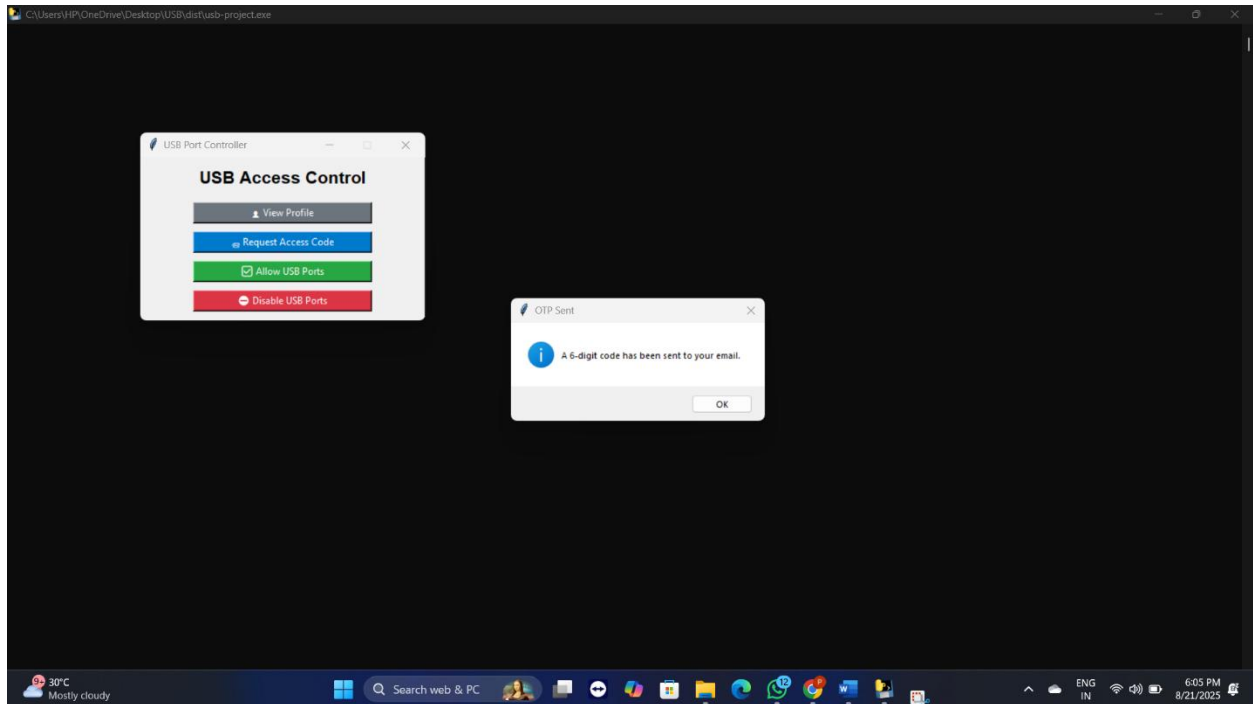
End-users confirmed the system was **easy to use**, messages were clear, and intrusion detection added a strong sense of accountability.

The project successfully passed acceptance testing, meeting **all functional and non-functional requirements**. The system was deemed **ready for deployment** in academic labs, corporate offices, and organizational environments where USB misuse is a threat.

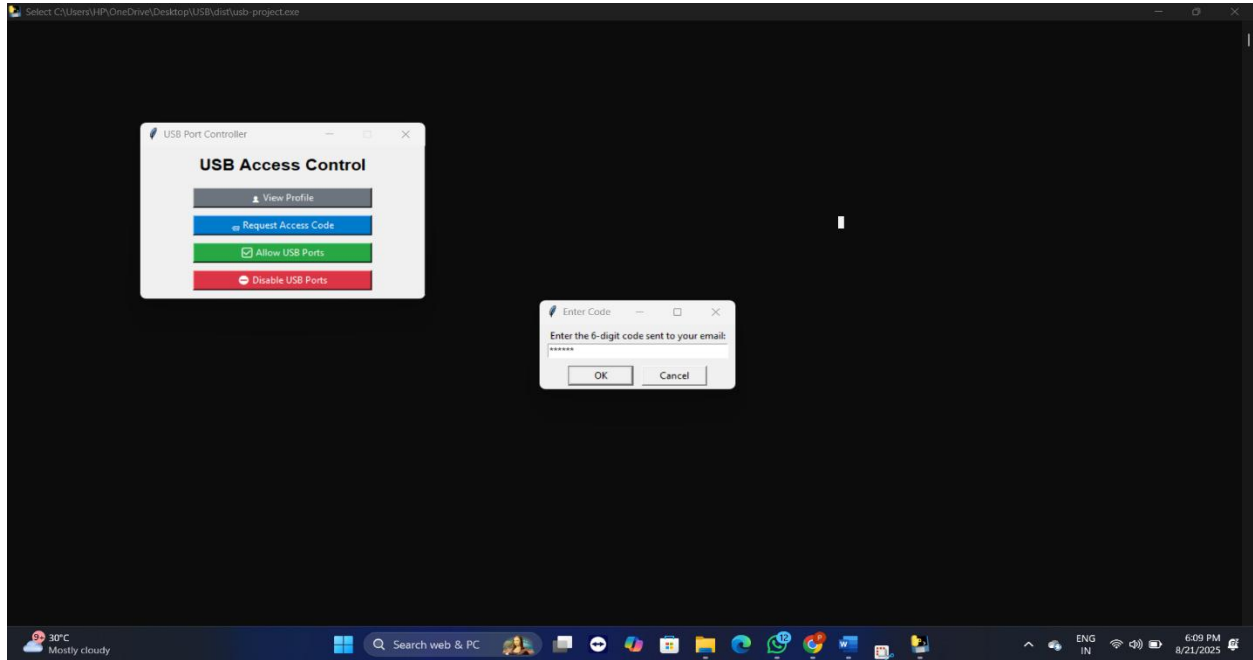
10. Execution Results:



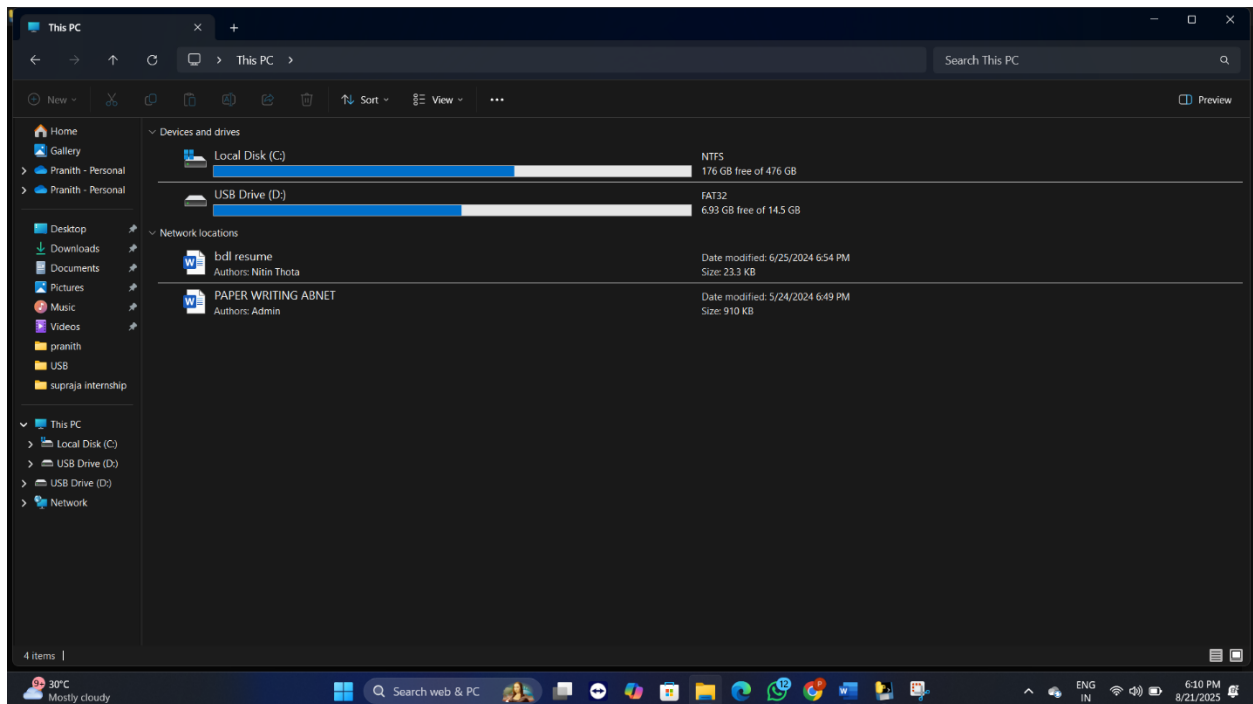
Click on the Request Access Code



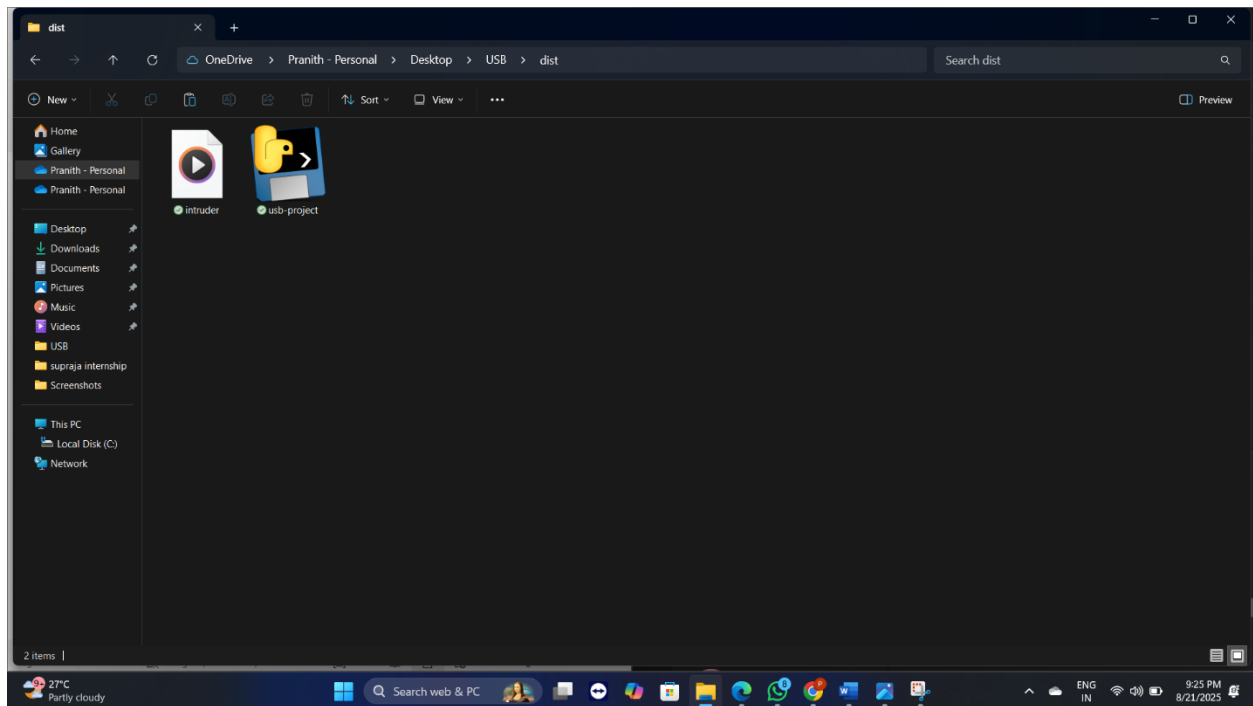
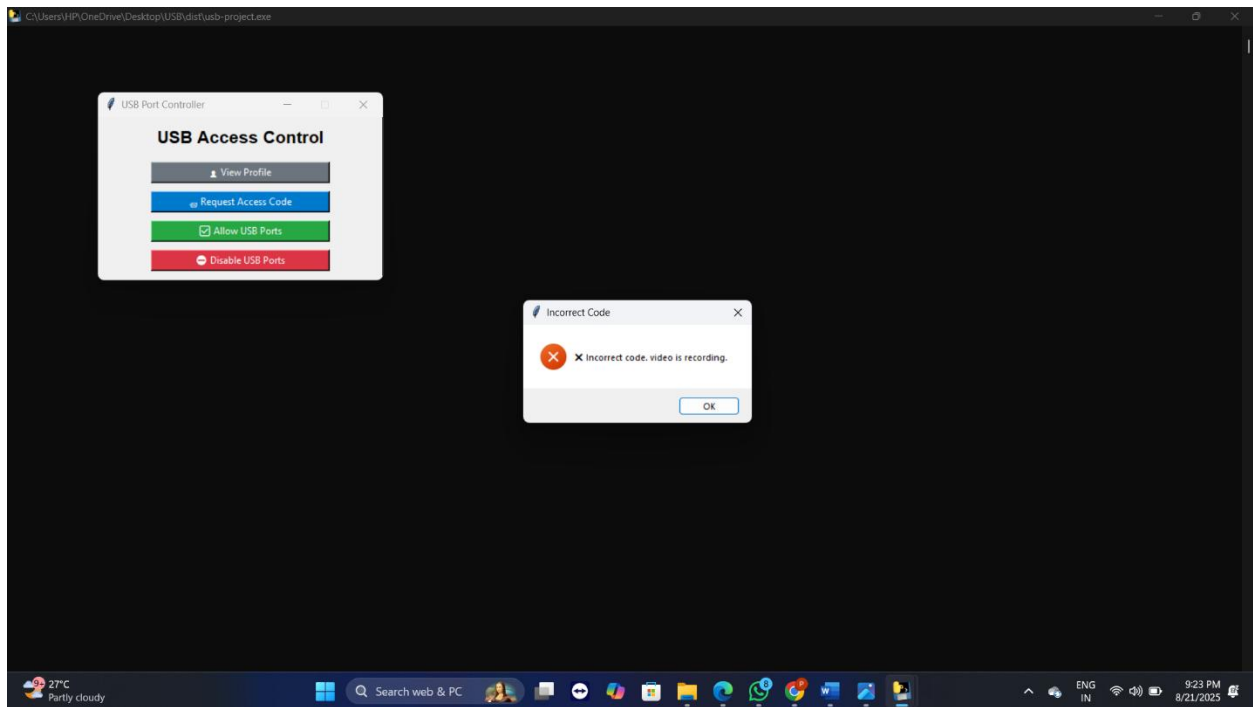
Click on the Enable button and enter the OTP and click on ok button.



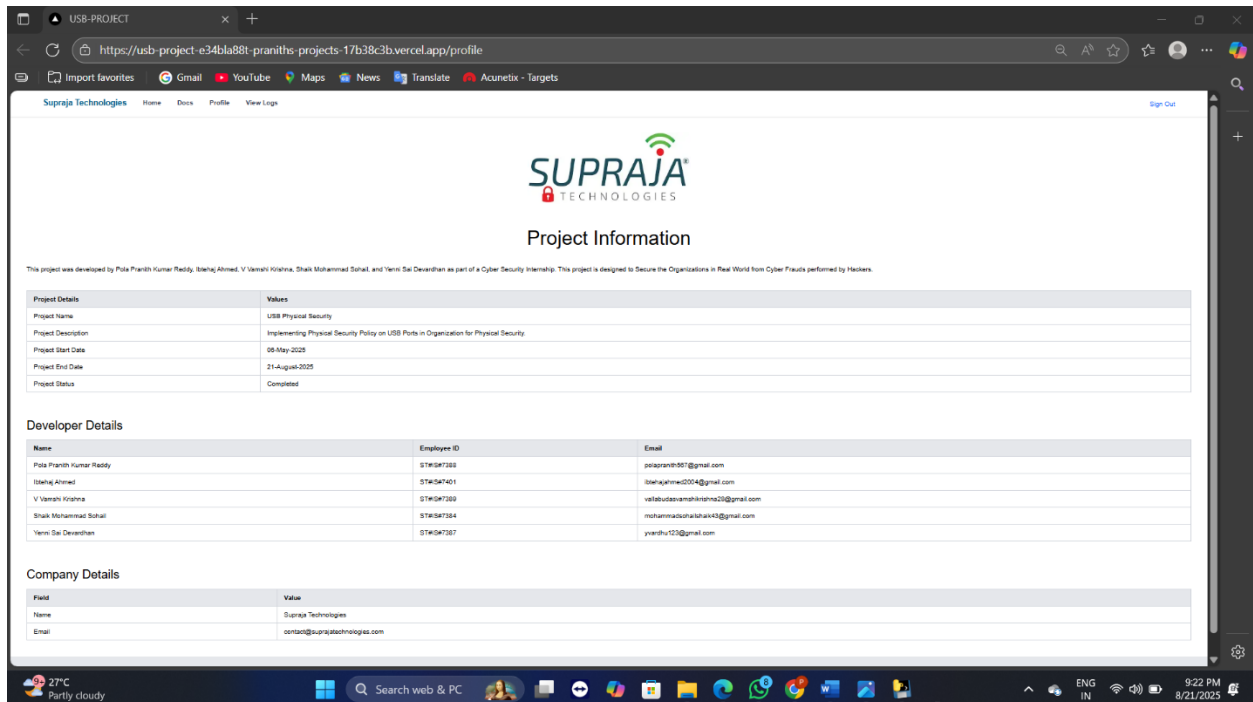
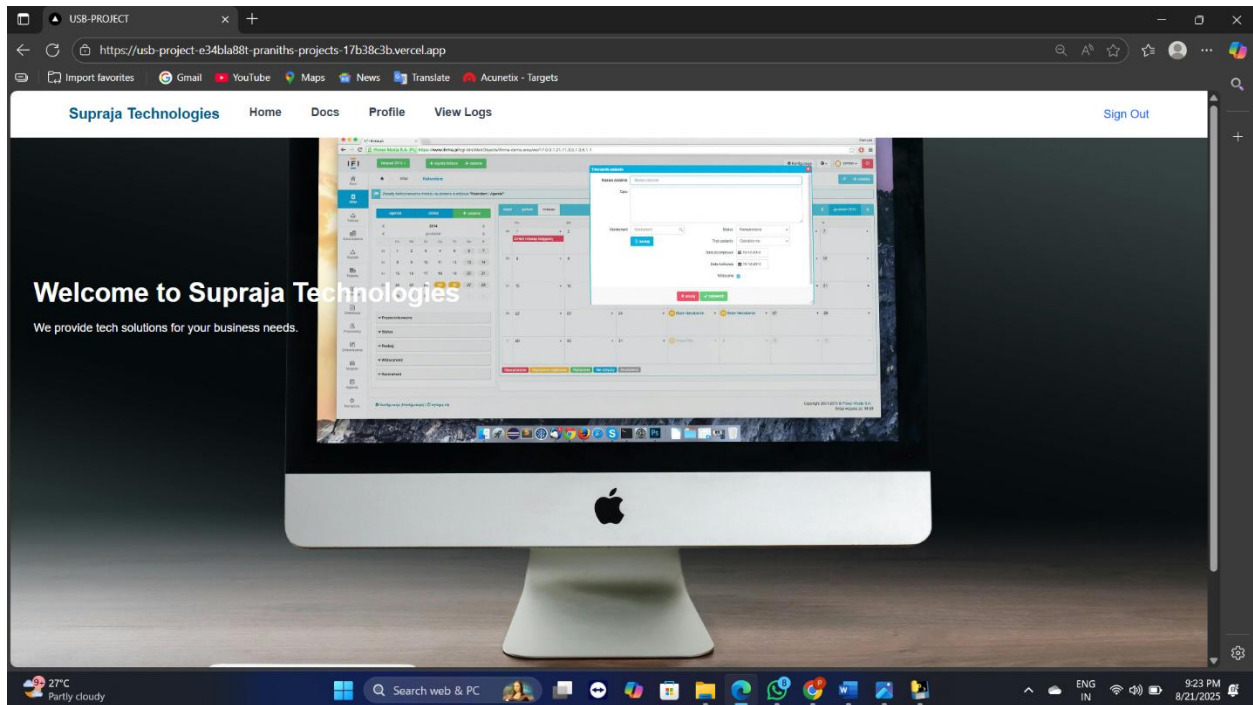
Then we can see a USB Drive in the File Manager.



When we enter the wrong OTP then it will record the video for 5 sec.



When we click on view profile button



USB-PROJECT x +

https://usb-project-e34b1a88t-praniiths-projects-17b38c3b.vercel.app/view-logs

Import favorites Gmail YouTube Maps News Translate Acunetix - Targets

Supraja Technologies Home Docs Profile View Logs Sign Out

Delete Logs

Timestamp	Code Generated	Email Sent	Disabled Ports	Enabled Ports	Video Recorded
2025-08-21 16:06:03	True	True	True	False	False
2025-08-21 16:06:03	True	True	True	False	Video saved to intruder.avi
2025-08-21 18:04:18	True	True	False	False	Video saved to intruder.avi
2025-08-21 18:08:46	True	True	False	True	False
2025-08-21 19:25:56	True	True	False	False	Video saved to intruder.avi

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27°C Party cloudy Search web & PC ENG IN 9:22 PM 8/21/2025

11. Conclusion

The USB Physical Security System was designed and developed to address the growing risks associated with unauthorized USB device usage. In many organizations, educational institutions, and workplaces, USB ports represent both a productivity tool and a security vulnerability. Malicious insiders or unauthorized users can exploit open USB ports to steal confidential data, inject malware, or compromise system integrity. This project offers a robust solution by introducing OTP-based authentication, intrusion detection, and logging mechanisms to ensure controlled and accountable access.

Through the implementation of modules such as Email OTP Verification, USB Port Enabling/Disabling, Intrusion Detection via Webcam, Logging, and a GUI Controller, the system combines authentication, monitoring, and reporting into a single integrated framework. The OTP mechanism ensures that only authorized users with access to a verified email account can enable or disable USB ports. Intrusion detection further strengthens security by recording unauthorized attempts, ensuring accountability and deterrence.

The system design followed a structured methodology, including feasibility studies, data flow modeling, UML diagrams, and ER modeling, to guarantee scalability and maintainability. Extensive testing methods such as unit testing, integration testing, and acceptance testing validated the reliability of each module, confirming that the system performs as expected under real-world conditions.

In conclusion, this project successfully delivers a multi-layered security system that protects USB ports from unauthorized access. It integrates user authentication, real-time monitoring, and systematic logging, providing both prevention and detection measures. The solution is practical, cost-effective, and adaptable for deployment in academic labs, corporate offices, and government organizations where USB misuse is a serious concern. Ultimately, the project demonstrates how combining simple authentication with vigilant monitoring can significantly strengthen physical and data security in computing environments.